Proof That Displacement Interpolation Gives Wasserstein Geodesics

Assume $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ and that the optimal transport for the quadratic cost is induced by a Brenier map

$$T : \mathbb{R}^d \to \mathbb{R}^d, \qquad T_\# \mu = \nu.$$

For the quadratic cost, define

$$T_t(x) = (1-t)x + tT(x), \qquad t \in [0,1],$$

and

$$\mu_t = (T_t)_\# \mu.$$

This curve moves each particle along the straight line from x to $T(x)$.

Claim

The curve $(\mu_t)_{t\in[0,1]}$ is a constant-speed geodesic:

$$W_2(\mu_s, \mu_t) = |t-s|\, W_2(\mu, \nu), \qquad s, t \in [0,1].$$

Proof

Fix $0 \le s < t \le 1$. Consider the coupling

$$\gamma_{s,t} = (T_s, T_t)_\# \mu \in \Pi(\mu_s, \mu_t).$$

This coupling pairs the position at time s and the position at time t of the same original particle x. For each such particle,

$$|T_t(x) - T_s(x)| = |(t-s)(T(x) - x)| = |t-s|\,|T(x) - x|.$$

Using $\gamma_{s,t}$ as an admissible transport plan gives

$$W_2^2(\mu_s, \mu_t) \le \int_{\mathbb{R}^d \times \mathbb{R}^d} |a-b|^2 \, d\gamma_{s,t}(a,b) = |t-s|^2 \int_{\mathbb{R}^d} |T(x) - x|^2 \, d\mu(x).$$

Since T is optimal from μ to ν,

$$\int_{\mathbb{R}^d} |T(x) - x|^2 \, d\mu(x) = W_2^2(\mu, \nu).$$

Thus

$$W_2(\mu_s, \mu_t) \le |t-s|\, W_2(\mu, \nu).$$

It remains to prove the reverse inequality. Apply the same upper bound on the intervals $[0, s]$ and $[t, 1]$:

$$W_2(\mu, \mu_s) \le s\, W_2(\mu, \nu), \qquad W_2(\mu_t, \nu) \le (1-t)\, W_2(\mu, \nu).$$

By the triangle inequality,

$$W_2(\mu, \nu) \leq W_2(\mu, \mu_s) + W_2(\mu_s, \mu_t) + W_2(\mu_t, \nu).$$

Therefore

$$W_2(\mu, \nu) \leq sW_2(\mu, \nu) + W_2(\mu_s, \mu_t) + (1 - t)W_2(\mu, \nu).$$

Rearranging,

$$W_2(\mu_s, \mu_t) \geq |t - s| \, W_2(\mu, \nu).$$

Combining the upper and lower bounds gives

$$W_2(\mu_s, \mu_t) = |t - s| \, W_2(\mu, \nu).$$

This is exactly the constant-speed geodesic condition.

Interpretation

Euclidean geodesics move points along straight lines. Wasserstein geodesics move mass along optimal straight-line particle trajectories. This is why the curve $\mu_t = (T_t)_{\#}\mu$ is called displacement interpolation.